

Declaration

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner (see cover page) and I under the guidance of our supervisor (see cover page) in the 2024 academic year at the Department of Engineering Science and Biomedical Engineering, Faculty of Engineering, University of Auckland.
2. This report is not the outcome of work done previously.
3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.
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Inferring the Detailed History of a Geothermal Field Using Sparse Data and Machine Learning

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Abstract

Geothermal reservoir modelling allows for the health of geothermal reservoirs to be accurately determined, guiding management on safe fluid extraction rates that maximize the longevity of the reservoir. Mass flow measurements from these reservoirs feed in as inputs to these reservoir models but are sparsely recorded for individual geothermal wells. These reservoir models require densely recorded mass flow measurements to generate robust predictions but often don't have access to them. As such, the lack of these measurements translates to uncertain reservoir models that potentially are erroneous. Existing methods for imputing mass flows across these wells are tedious and potentially erroneous as geothermal experts manually calibrate models. Parameters are adjusted until the models produced align closely with the recorded mass flow data, and predictions are made. This process can take months and requires expertise to calibrate these models, which is expensive, both time and monetary wise.

Gaussian process regression offers an effective method for imputing mass flows by creating uncertainty intervals between recorded data. Correlations between data points drive the width of this uncertainty interval and, hence, the range of possible predictions. The kernel, commonly known as a covariance matrix, determines how correlation changes between these data points as time progresses. The sum of flow across wells feeding into a separator must match the densely recorded total flow. As such, these model implementations were modified to obey this constraint.

Using the same real-geothermal data, the Gaussian process regression and mathematical formulation approach are compared to each other and the expert's estimate. The Gaussian process regression approach successfully went through each of the recorded data points and sometimes captured the expert's estimate within its uncertainty interval. However, the mathematical formulation appeared more robust as this method matched the expert's estimates more closely, and its predictions oscillated less than those of the Gaussian process regression approach. Further research into this Gaussian process regression approach will likely make it an effective tool for mass flow imputation that can replace manual calibration methods.

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Chapter 1

Introduction

Geothermal energy is a widely used renewable energy source extracted from the Earth's core [1]. A geothermal reservoir, commonly called a geothermal resource, contains very hot rocks and fluids, with thermal energy inside these rocks and fluids [1]. Steam and hot water from the geothermal reservoir can be used directly for heating and cooling applications or electricity generation [2]. Direct heating and cooling applications, like geothermal heat pumps, are typically used for homes and other small-scale buildings [3]. Electricity generation, in contrast, is a large-scale operation in New Zealand, with companies such as Mercury Energy and Contact Energy owning geothermal fields that extract this energy to produce electricity [4]. The electricity is then sold to customers all around New Zealand, with customers ranging from individual households to large commercial businesses.

Generating electricity requires fluid to be extracted from the geothermal reservoirs using geothermal wells. As with any energy source, a finite amount of fluid can exist within these geothermal reservoirs, meaning reservoir management is essential when using these reservoirs for electricity generation. As geothermal energy is a renewable energy source, the energy used will be replaced by new energy after some time. By ensuring moderate extraction rates are used for the wells, the geothermal reservoir can remain healthy and thus maximize the field's longevity [5]. Without the proper management of these reservoirs, there's a potential that too much fluid will be extracted from the reservoirs, putting the reservoir at risk of early depletion.

Geothermal reservoir models can model a given reservoir using the relevant reservoir parameters and well measurements. By passing off the required inputs, the health and status of the reservoir can be determined at any time. As such, this can readily inform the geothermal operator whether the extraction rates are too high for a particular well. With many potential measurements existing, having a strong understanding of them is key to modelling a reservoir effectively.

1.1 Measurements

Mass flow, the amount of mass extracted from a reservoir by a well, is crucial for modelling processes like the pressure response inside the reservoir. These processes dictate the reservoir's health, so having access to densely recorded datasets that track mass flow is essential to accurate reservoir modelling. Unfortunately, mass flow measurements are sparsely recorded because steam flow measurements are often of more interest to geothermal operators since the steam flow generated corresponds to how much electricity is produced. To make matters more challenging, only the total steam flow measurements for a collection of geothermal wells feeding into a single separator are recorded frequently. Individual steam flow measurements taken at a well must be used with total steam flow measurements to impute missing steam flows and, thus, subsequent mass flows for the wells.

Consider Figure 1.1 below, illustrating a simple geothermal system. Three wells feed into a separator that separates the steam from the steam and water mixture. The steam from these three wells feeds into a generator that produces electricity. Steam feeding into the

generator is regularly recorded, but the individual wells' steam and mass flow measurements are not. Knowing the fluid extraction rates across all three wells would allow safe extraction rates to be determined through reservoir modelling, informing geothermal operators about their current rates. Consequently, these extraction rates could be readjusted by the geothermal operator so the longevity of the reservoir is maximized. For example, if the extraction rate of well A increased, then, with dense mass flow data, it could be determined how safe this new extraction rate is and whether wells' B and C extraction rates would need to be adjusted. However, not having access to this dense data creates uncertainty surrounding extraction rates.

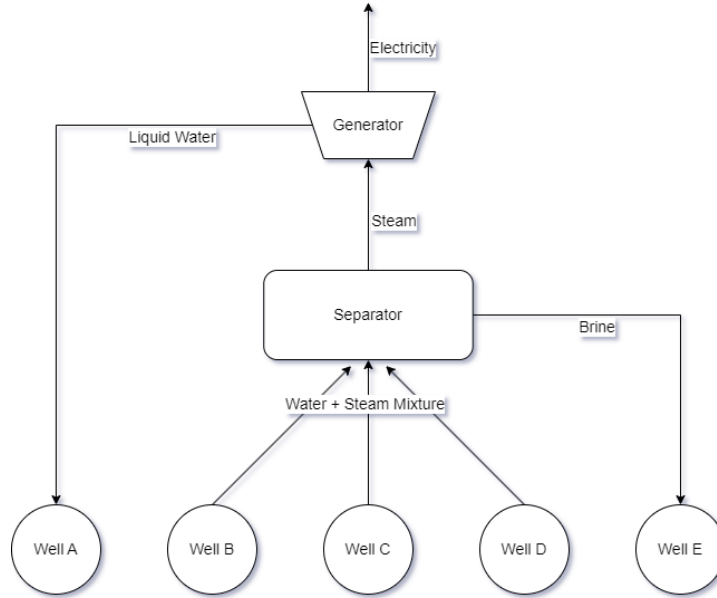


Figure 1.1: A simple geothermal system with five wells, a separator, and a generator. Adapted from [6]

These geothermal systems can become large, and thus, the lack of individual well measurements, especially mass flow measurements, makes decisions regarding the geothermal field challenging. Wairakei geothermal field, for example, has over 290 wells [7]. Geothermal experts currently use manual calculations to infer the mass taken from wells in a geothermal field. This process is time-consuming because it uses an ad-hoc approach where parameters are guessed and tried. Essentially, it is a guess-and-check method. Experts in the geothermal space can often spend months determining feasible and likely combinations of these parameters to make the necessary mass flow inferences. It is time-consuming, and there is a high potential for producing erroneous results as this method introduces lots of uncertainty, especially with such a large parameter space. As such, these current practices need to be improved as spending months to get a potentially erroneous solution with unquantified uncertainty is costly both monetary and time-wise.

1.2 Literature Review

Geothermal reservoir modelling allows the geothermal operator to monitor, manage, and expand reservoirs they use for electricity generation. Many aspects go into reservoir modelling, as parameter knowledge and high-quality data are needed to generate good results. As these models represent the fluid flow within the reservoir [8], generating accurate models is key to successfully managing a geothermal field. Geothermal wells extract mass from reservoirs at a significant rate and do this for a while, so having a dense dataset containing the extraction history of wells is crucial for reservoir modelling. Unfortunately, as discussed earlier, dense data, while being an important aspect of reservoir modelling, is not available. This section will explore methods and techniques that may prove useful in solving this sparse data problem while providing analogous problems that have successfully applied these techniques.

1.2.1 Gaussian Process Regression (GPR)

GPR is a powerful non-parametric model and stochastic process used in supervised machine learning problems [9]. It can be used to formulate a Bayesian framework for both regression and probabilistic classification problems where many, possibly even infinite, random variables are involved. Initially, a prior distribution is formulated where a distribution is fit across a function without considering additional information. This prior distribution uses the mean and covariance function to define a Gaussian process [10]. Using this allows for a function to be fit without any data observations and thus provides some levels of uncertainty before adding information to the problem. Rasmussen [9] demonstrates the prior distribution being fitted using a Gaussian Process on three functions. There are no data points, and the functions are created using fitted or estimated parameters, as shown in Figure 1.2 below.

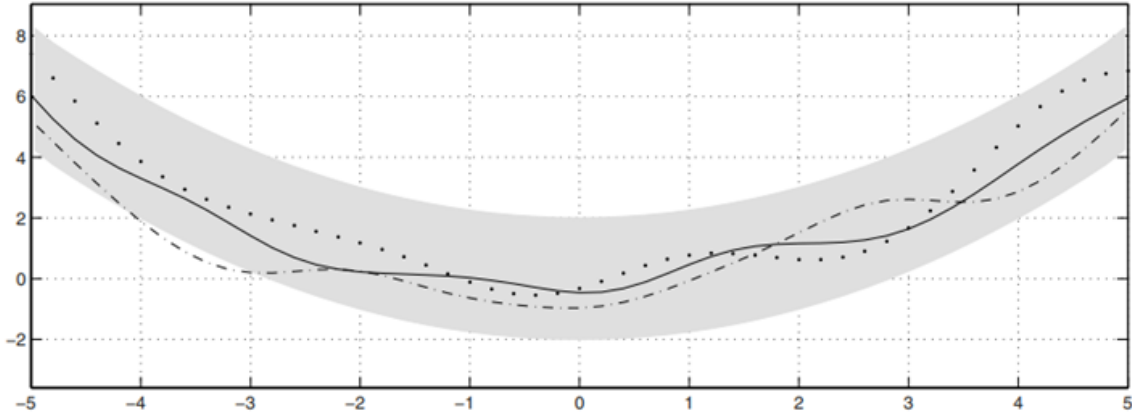


Figure 1.2: Example of a Gaussian Process applied to 3 random functions creating the prior distribution. Adapted from [9]

Once the prior distribution is defined, the posterior distribution can be defined, which uses both the prior distribution and existing data observations to create a more accurate function. Using a set of equations, the mean and covariance of the posterior distribution can be determined. Variance in the posterior distribution is less than the prior because it takes the prior variance and subtracts a positive term from it, which depends on the training data. In Figure 1.3, the posterior distribution using the prior distribution in Figure 1.2 and the training data is shown.

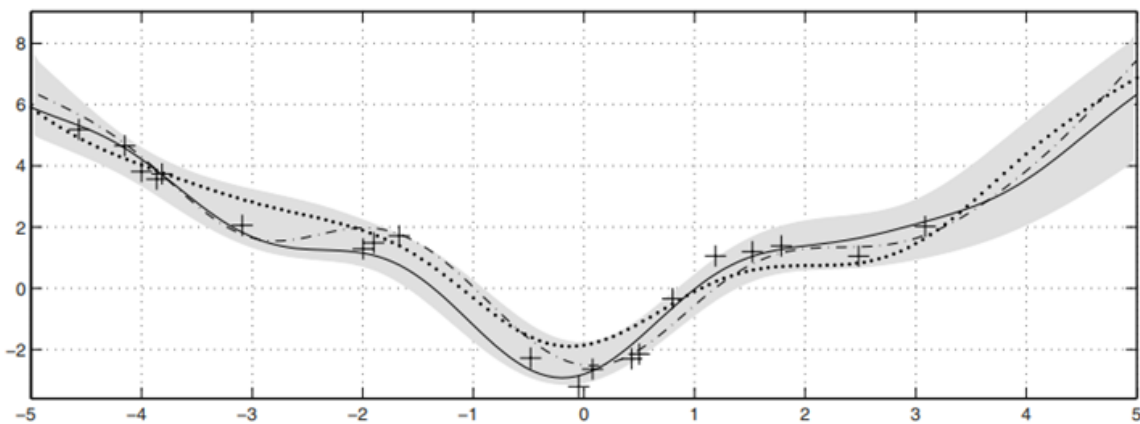


Figure 1.3: Example of combining the prior distribution and training data to create the posterior distribution. Adapted from [9]

As seen from Figures 1.2 and 1.3, there are uncertainty intervals for the prior and posterior distribution. Using these uncertainty intervals, realizations for the target output variable

can be drawn using the corresponding input variables. Multiple realizations can be drawn to make an informed decision about the most likely value for this particular input.

While GPR hasn't been used in the geothermal context before, it has the characteristics to be a useful solution method. The geothermal mass flow problem introduces uncertainty due to the extremely sparse data. As such, having a method that can first fit a model with this sparse data and second provide a range of realistic realizations under this model would be invaluable. Uncertainty intervals created from a GPR model would allow mass flow realizations to be created and, hence, would naturally tackle the uncertainty surrounding this problem.

GPR has been applied to other unrelated problems that deal with sparse data. For example, GPR was used to predict the Young's modulus of oxide glasses using sparse datasets [11]. The research demonstrated that GPR performed well for this prediction problem and even outperformed neural networks, which are known to be robust machine learning algorithms. Regardless of the unrelated field, the successful application of GPR for solving sparse data problems strengthens the idea of using GPR for the geothermal sparse data problem.

1.2.2 Ensemble Methods

Ensemble methods aim to combine multiple models to improve accuracy rather than just using a singular model [12]. Three common types of ensemble methods exist, with them being bagging, boosting, and stacking methods [12]. Broadly speaking, there are two distinct categories of ensemble methods: sequential ensemble techniques and parallel ensemble techniques. Sequential ensemble techniques build everything in series, meaning each learner depends on the others. Parallel-based methods, however, build everything simultaneously, which promotes independence among different learners.

Bagging methods involve three key steps, as outlined by Leo Breiman [13]. These three steps are bootstrapping, parallel training, and aggregation. The first step, bootstrapping, takes the original data and samples, with replacement, from this data until a sample of the same size as the original data is created. This process is repeated until a required number of complete samples are created. The second step, parallel training, trains these samples using weak learners. Finally, the third step, aggregation, uses the predictions drawn from each sample to determine the average or appropriate class, whichever one is relevant to the problem. Boosting methods iteratively combine weak classifiers to create stronger ones [14]. Various boosting methods exist, but the baseline is each iteration focuses on fixing the shortcomings of the previous iteration, so eventually, a strong classifier will be established. Common boosting methods are adaptive boosting, gradient boosting, and extreme gradient boosting. Stacking methods use multiple models to create a final prediction. First, predictions are made using multiple base models [15]. These predictions are fed to the final model, which generates the final prediction. Figure 1.4 illustrates the concept of stacking concisely.

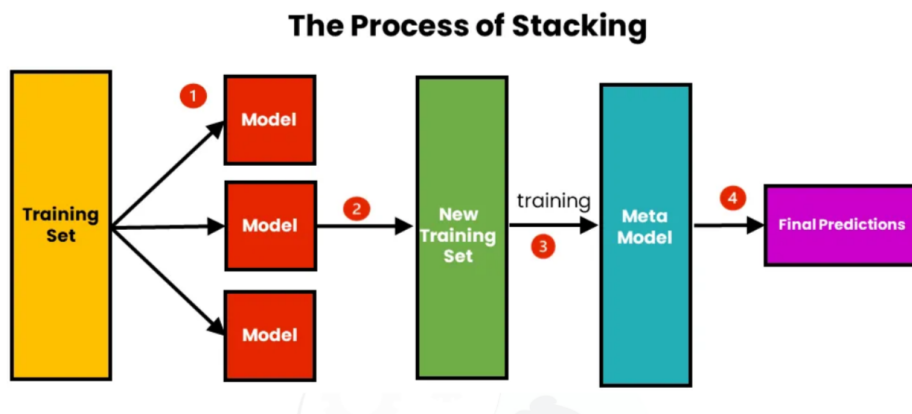


Figure 1.4: Illustration of how stacking works as an ensemble method. Adapted from [16]

These ensemble methods are highly applicable to solving this sparse data problem. By applying an appropriate ensemble method, reasonable predictions for missing mass flow could be readily made. Ensemble methods such as random forests, gradient boosting, and adaptive boosting could be used to determine likely mass flow values. The average mass flow, specified quantile range, most likely mass flow value, and many more values could be determined from these ensemble methods. As these are ensemble methods, predictions from various ensembles could deliver a robust method for predicting mass flow and thus inferring the geothermal history field.

Ensemble methods haven't been used for interpolating missing mass flow values in sparsely recorded geothermal problems. However, these ensemble methods have been used at a university for calibrating geothermal models [17]. Their problem focused on finding a good set of parameters to define the system accurately, known as model calibration. It was demonstrated that ensemble methods could produce accurate estimates relatively efficiently [17]. These ensemble methods could be further extended to solve for missing mass flow values rather than solving a model calibration problem.

1.2.3 Physics-Informed Machine Learning (PIML)

Physics-informed machine learning (PIML) is a tool used in machine learning-focused problems. It combines machine learning techniques with physical processes to enhance the models produced [18]. Courtesy of Max Marian and Stephan Treml [19], a concise image displaying the relationship between data quantity and physics applied to the problem is presented below.

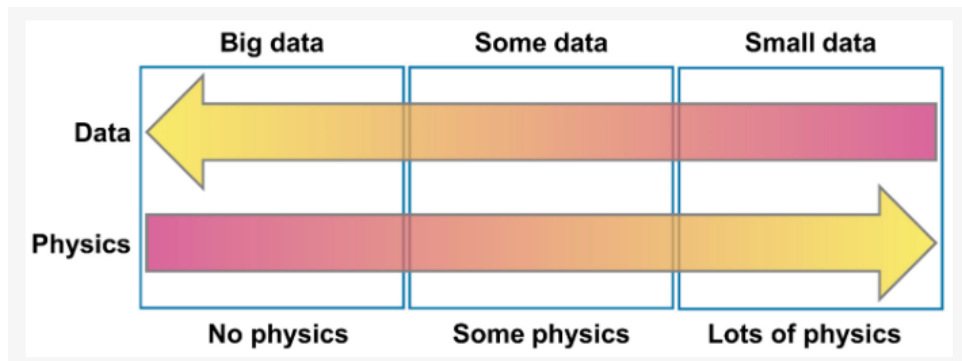


Figure 1.5: Relationship between data and physics when using PIML. Adapted from [19]

Figure 1.5 demonstrates the idea that, with increased physics, the need for as much data decreases. This allows suitable models to be created to solve sparse data problems, assuming enough physics is added. While adding more physics is often appropriate, it's important not to overconstrain the problem, as this may limit the model's ability to generate reasonable results.

Commonly, PIML is built using neural networks (NN), which are some of the most complex machine learning algorithms. The biggest strength of using NN is that they are quick to yield results due to their structure and, with good training, generate good predictions [18]. However, as with all NN, they require broad data to generalize well and produce good predictions [18]. This may not be possible in sparse data situations as there isn't enough data to produce a high-quality NN to use for PIML.

Often, when solving a geothermal problem, there is a lot of physics that needs to be carefully considered. Various things like permeability, porosity, and pressure are involved within a geothermal reservoir. As such, theoretically, introducing additional known values into the problem makes for more constrained and robust models and, therefore, can produce better models.

PIML is a popular technique for tackling geothermal problems like reservoir management. Recently, a research paper detailed the use of PIML for reservoir management of enhanced

geothermal systems (EGS) [20], which are reservoirs that have been enhanced or created through hydraulic simulation [1]. Currently, reservoir simulation is used to monitor and manage EGS, which is costly and complex. An alternative to this approach using PIML was proposed, which would take advantage of the naturally large level of physics involved in geothermal reservoirs to manage EGS in a more cost-friendly manner. While the sparse data problem involves estimating missing mass flow values rather than reservoir management, the same principles could be applied to solve the problem. Developing a deep understanding of the geothermal field would help create a detailed model that would predict mass flow values fairly accurately, even with sparse data, as the physics could compensate for this.

While this method is promising, it carries challenges surrounding the physics. Geothermal reservoirs are complex and unique, meaning each geothermal reservoir would require custom parameters to be set, which is a very challenging task for large geothermal fields. Secondly, expert domain knowledge would be needed to infer and set the relevant physical parameters accurately.

1.2.4 Mathematical Formulation

Assuming the problem is understood well enough and can be parameterized, a relevant mathematical model can be formed to solve directly for mass flow. This would involve solving a large system of linear equations using matrix and vector operations. Unfortunately, this hasn't been implemented in a geothermal context so there is little literature surrounding its application for this sparse data problem. However, César Massri demonstrated in his 2016 paper how a linear algebra approach could solve sparse systems [21]. This linear algebra approach utilized eigenvalues, eigenvectors, and coefficient matrices, which could be adapted to the geothermal context. As linear algebra is very well understood and easy to use, this may be a suitable method for imputing missing mass flows.

1.2.5 Regression Based Methods

One robust method for imputing missing data is employing regression-based methods like linear, LASSO, and ridge regression. These methods are simple to understand and work well when there is a correlation between the missing value and other variables [22]. They can be readily implemented using common Python packages like Seaborn and Scikit-Learn and offer much flexibility. The sparse nature of the geothermal field data means using a LASSO regression model could be an appropriate way forward. The datasets will likely be large yet sparse due to many missing observations, so LASSO regression could be an effective tool to build a robust imputation model that utilizes only a small subset of available variables [23]. These regression methods are not without risks, however, as they all require external data for validation and can be at risk of overfitting [24]. Validation of the model may not be possible due to an already extremely limited set of available data collected by geothermal operators.

1.3 Research Scope

This research report will focus on the application and effectiveness of using Gaussian process regression (GPR) models to impute missing mass flows for individual wells using the sparse data provided. Synthetic data will be used initially to demonstrate the feasibility of these GPR models. Afterwards, the GPR models will be tested on real-geothermal data from Contact Energy. Daniel Clark, my project partner, has focused on implementing a mathematical formulation approach [6] to solve this imputation problem. Using a selected subset of the real-geothermal data, the effectiveness of the GPR and mathematical formulation approach will be compared. Creating these GPR models will devise a robust way for missing mass flows to be predicted across wells in geothermal fields. Geothermal operators and modellers would highly benefit from such a solution, as the history of a geothermal field could be inferred.

Chapter 2

Methods

Before applying Gaussian process regression (GPR) to this sparse data problem, an in-depth mathematical description of GPR is provided. This mathematical description aims to provide insight into how GPR is working and solving the problem it's presented with. Relevant libraries and tools used for solving the provided problem are also discussed.

2.1 Gaussian Process Regression (GPR)

Gaussian process regression (GPR) is a clever machine-learning method that uses complex mathematics. A GPR model is made up of Gaussian processes, which are described to be a finite collection of joint Gaussian distributions defined by random variables [9]. Consider first a Gaussian distribution in the form

$$X \sim \mathcal{N}(\mu, \sigma^2) \quad (2.1.1)$$

where X is normally distributed with a mean of μ and a variance of σ^2 where σ is the standard deviation. Naturally, this Gaussian distribution is of one-dimensional space and thus only models one random variable. As a Gaussian process requires multiple random variables, this Gaussian distribution must be extended to include multiple random variables. A joint Gaussian distribution, commonly known as a multidimensional Gaussian distribution, depends on multiple continuous random variables [25]. Equation (2.1.1) can be rewritten now as

$$X \sim \mathcal{N}(\mu, \Sigma) \quad (2.1.2)$$

or

$$X \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_d \end{bmatrix}, \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_{dd} \end{bmatrix} \right) \quad (2.1.3)$$

where X is a normally distributed random vector, μ is a vector of means, and Σ is a covariance matrix. As such, it can be seen that the joint Gaussian distribution is defined by its mean vector and covariance matrix.

A Gaussian process takes this joint Gaussian distribution definition and extends it further by making it such that both its mean and covariance function define it [9]. This can be expressed as

$$f \sim \text{GP}(m, k) \quad (2.1.4)$$

where function f is distributed as a Gaussian process with a mean of m and covariance of k [9]. It can be imagined that f is a distribution over a function, and hence, samples can be repeatedly drawn to produce the prior distribution [9]. This prior distribution occurs in the absence of training data and is used to gain some insight into the potential distribution of the function itself.

Incorporating data into the prior distribution creates the posterior distribution. Consider f to be function values corresponding to the training cases and correspondingly f_* to be the unknown function values for the test cases. Then, the joint distribution that contains both the test and training cases of interest can be expressed as

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu} \\ \boldsymbol{\mu}_* \end{bmatrix}, \begin{bmatrix} \Sigma & \Sigma_* \\ \Sigma_*^\top & \Sigma_{**} \end{bmatrix} \right) \quad (2.1.5)$$

where μ and μ_* are the training and test means, respectively, and Σ represents the training set covariance, Σ_* represents the training-test set covariances, Σ_{**} represents the test set covariances, and $\Sigma_*^\top$ is the training-test covariance matrix transposed [9]. It's possible to find an expression for $f_* | f$ using equation (2.1.5) by using the property

$$\left\{ \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix}, \begin{bmatrix} A & C \\ C^\top & B \end{bmatrix} \right) \right\} \implies \mathbf{x} | \mathbf{y} \sim \mathcal{N} (\mathbf{a} + CB^{-1}(\mathbf{y} - \mathbf{b}), A - CB^{-1}C^\top) \quad (2.1.6)$$

which maps the joint distribution to a conditional expression. Applying equation (2.1.6) to equation (2.1.5) gives

$$\mathbf{f}_* | \mathbf{f} \sim \mathcal{N} (\boldsymbol{\mu}_* + \Sigma_*^\top \Sigma^{-1}(\mathbf{f} - \boldsymbol{\mu}), \Sigma_{**} - \Sigma_*^\top \Sigma^{-1} \Sigma_*) \quad (2.1.7)$$

which is the posterior distribution expression for the test cases of interest. Translating these results back to posterior processes gives these two key equations

$$m_{\mathcal{D}}(x) = m(x) + \Sigma(X, x)^\top \Sigma^{-1}(f - m) \quad (2.1.8)$$

$$k_{\mathcal{D}}(x, x') = k(x, x') - \Sigma(X, x)^\top \Sigma^{-1} \Sigma(X, x') \quad (2.1.9)$$

which represent the posterior mean, $m_{\mathcal{D}}$, and the posterior covariance matrix, $k_{\mathcal{D}}$. In this case, x is just an input to the relevant function. As $m_{\mathcal{D}}$ and $k_{\mathcal{D}}$ are defined, the corresponding expression for the function conditioned on the posterior can be defined for a Gaussian process as

$$f | \mathcal{D} \sim \text{GP}(m_{\mathcal{D}}, k_{\mathcal{D}}) \quad (2.1.10)$$

where equation (2.1.10) combines elements from equations (2.1.4), (2.1.7), (2.1.8), and (2.1.9). All the pieces surrounding the equation are known, and hence, the posterior process and distribution can be used to generate predictions for the test cases.

To better understand the prior and posterior distributions defined above, Figure 2.1 is presented, which displays the prior and posterior distributions for a function. Figure 2.1 demonstrates the impact conditioning can create when just a few training data points exist. The distribution becomes a lot more refined, and hence, the uncertainty decreases, and the overall quality of the model improves.

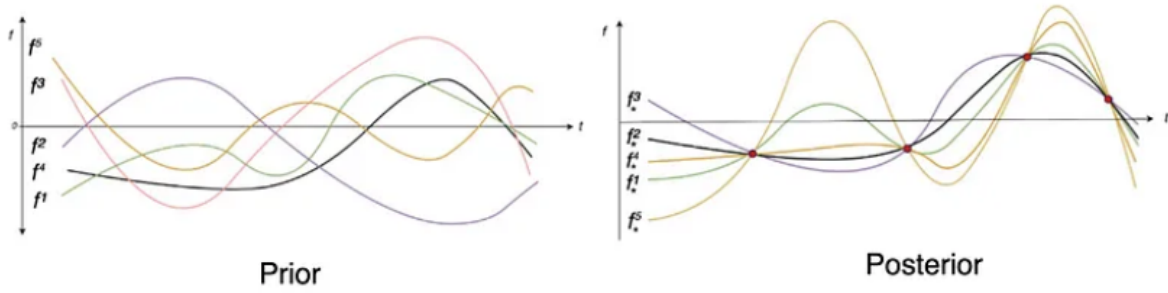


Figure 2.1: A simple demonstration of prior and posterior distributions.
Adapted from [15]

Everything discussed before relies on having a suitable prior distribution to generate the posterior distribution. Often, this isn't the case where the potential behaviour of the function itself isn't known at all. A prior distribution can't be fit in this case, meaning the GPR needs to be fit using the limited training data points supplied directly. If appropriate mean and covariance functions can be chosen for a Gaussian process, then a well-suited GPR model can be created.

The mean and covariance functions are defined by their parameters. For example, a linear equation would be defined by its slope and intercept. These parameters could be treated as hyperparameters that need to be optimized. If θ_m contains the hyperparameters for the mean function and θ_k contains the covariance function hyperparameters, then θ can be seen as the complete hyperparameter set that contains θ_m and θ_k .

The log marginal likelihood is used to determine the optimal set of hyperparameters by finding the maximum [26]. If the negative log marginal likelihood is taken, minimizing the function will give the same optimal hyperparameter set. The negative log marginal likelihood will be used in this report. Mathematically, the log marginal likelihood can be given as

$$L = \log p(y | x, \theta) = -\frac{1}{2} \log |\Sigma| - \frac{1}{2} (y - \mu)^\top \Sigma^{-1} (y - \mu) - \frac{n}{2} \log(2\pi) \quad (2.1.11)$$

where the optimal set of hyperparameters can be determined by taking the partial derivatives of the log marginal likelihood function with respect to θ_m and θ_k , respectively [9]. These partial derivatives are given as

$$\frac{\partial L}{\partial \theta_m} = -(y - \mu)^\top \Sigma^{-1} \frac{\partial \mu}{\partial \theta_m} \quad (2.1.12)$$

$$\frac{\partial L}{\partial \theta_k} = \frac{1}{2} \text{trace} \left(\Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_k} \right) + \frac{1}{2} (y - \mu)^\top \frac{\partial \Sigma}{\partial \theta_k} \Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_k} (y - \mu) \quad (2.1.13)$$

for θ_m and θ_k , respectively. Equation (2.1.11) can be broken down into three distinct terms.

Table 2.1: Decomposition of the log marginal likelihood function into its components. Complete equation is (2.1.11)

Term	Expression
Complexity Penalty Term	$-\frac{1}{2} \log \Sigma $
Negative Quadratic Term	$-\frac{1}{2} (y - \mu)^\top \Sigma^{-1} (y - \mu)$
Log Normalization Term	$-\frac{n}{2} \log(2\pi)$

Table 2.1, breaks down the log marginal likelihood function into three terms, emphasising the purpose each term has. The complexity penalty term acts as a measure of the model's complexity [9]. This parameter plays a big role in ensuring the model doesn't underfit by being too simple or doesn't overfit by making it too complex. The negative quadratic term is the data-fit measure term that ensures the provided fit matches the data to an acceptable degree [9]. The log normalization term is not interesting as it has little significance [9]. Considering the first two terms discussed, it could even be seen as a tradeoff between the quality of fit and the complexity of this fit.

2.2 Libraries & Tools

Various libraries and tools were used to develop usable GPR models to solve this sparse data problem. Python was the programming language used to develop and implement the GPR models. This language was selected because of its popularity, high versatility, existing IDEs that allow for extensive debugging approaches, and it is free. Very common libraries that won't be discussed in large detail were NumPy, Matplotlib, and pandas. Numpy was essential for manipulating and utilizing arrays and matrices, common data structures, and calculating standard statistics. Matplotlib performed the plotting functions, such as plotting the many graphs developed during the research. The Python library pandas allowed CSV files to be read and supplied the GPR models with the required data points. Scikit-Learn is a machine-learning library for Python. It contains many machine-learning algorithms such as linear models, decision trees, and, of course, Gaussian processes. It can generate Gaussian processes for regression and classification purposes, but this research utilized the regression purpose. The advantages of using Scikit-Learn are that it's accessible, easy to use, free, and well-integrated with Python. Unfortunately, Scikit-Learn is also not as powerful as other GPR-focused packages like GPy and GPflow.

2.2.1 ccandu

A key tool for producing conditional GPR models is called ccandu, pronounced as "can do", which is an in-house custom library built by the Modelling Uncertainty Data Group at the University of Auckland [27]. It's a highly specialized tool that utilises conditional random variables by combining and modifying them. Corresponding posterior distributions that satisfy given summative constraints could be readily drawn due to the conditional capabilities of this tool. Using ccandu offers a flexible and customisable approach to implementing the GPR model, as there is plenty of freedom to define aspects of the GPR model itself. As detailed later on in the report, ccandu replaces Scikit-Learn as the library to use for generating GPR models due to the reasons described and overall better performance.

Chapter 3

Results

The implementation of GPR on problems of increasing complexity is outlined, along with the quality of their predictions. Synthetic data is initially used to fit and test GPR models. Initially, a simple problem is solved to generate a proof of concept. Afterwards, more challenging problems are solved where assumptions are relaxed further. Later on, real-geothermal data is used for the GPR models to observe the performance of these current methods on problems that are likely to be encountered by geothermal operators. Finally, it will be determined whether there's a substantial difference between the GPR and mathematical formulation method and which ones, if any, perform better for the varying problems.

3.1 Independent GPR Processes

The simplest case of implementing a GPR model for this problem is considering only two wells in the geothermal field, and these two wells are always flowing. Scikit-Learn's implementation of a GPR model requires an appropriate kernel, interchangeably referred to as a covariance matrix, to be specified. The earliest implementation of this GPR model uses a Rational Quadratic kernel. Mathematically, this kernel is expressed as

$$k(x_i, x_j) = \left(1 + \frac{d(x_i, x_j)^2}{2\alpha l^2}\right)^{-\alpha} \quad (3.1.1)$$

where α is the scale mixture parameter, l is the length scale, and $d(x_i, x_j)$ is the Euclidean distance between x_i and x_j . Using this, two GPR models are fitted for the two wells using $l_{\text{bounds}} = (0.1, 100)$ and $\alpha = 10^{-10}$. The sum of steam flows across both wells should sum to the total steam flow, however, this condition is not enforced as the GPR models are fitted independently. Figure 3.1 displays the results of these two separate GPR models.

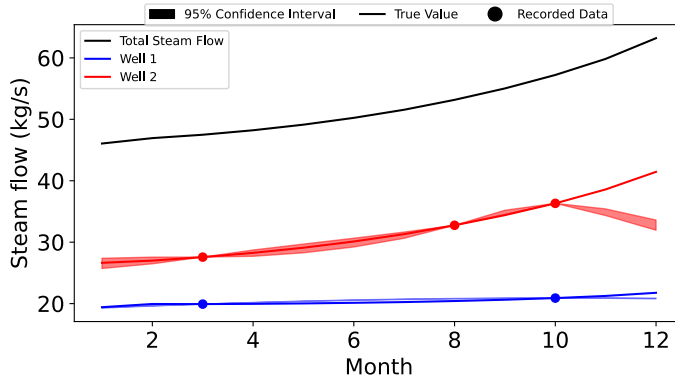


Figure 3.1: Two independent GPR models for two wells that are always flowing over 12 months

The GPR models for both wells capture the true steam flow values within their uncertainty intervals up to month 10. After this, the uncertainty bands fail to capture the steam flow values for months 11 and 12 as the predicted steam flow for both wells drops sharply when it should instead rise. This suggests the GPR model struggles with the extrapolation of steam flow values, which is a very common issue with machine learning algorithms. While the uncertainty intervals did capture the true steam flow values up until month 10, they were far too narrow and indicated a high level of confidence in the model created. With such sparse data and numerous potential applications, widening the uncertainty intervals would create a more flexible model that would better capture the broad range of possible values for missing steam flows. Adjusting the kernel by weakening the correlations between recorded steam flows, as well as introducing a scaling factor, would adjust the width of the produced uncertainty intervals. A Matern kernel was also used to see whether this generalized kernel was potentially better suited than the RationalQuadratic kernel. The Matern kernel is given below and can be expressed as

$$k(x_i, x_j) = \frac{1}{\Gamma(\nu)2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{l} d(x_i, x_j) \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{l} d(x_i, x_j) \right) \quad (3.1.2)$$

where ν controls the smoothness of the learned function, l is the length scale, $d(x_i, x_j)$ is the Euclidean distance between x_i and x_j , K_ν is the modified Bessel function, and $\Gamma(\nu)$ is the gamma function. The refitted GPR models using the RationalQuadratic kernel used $l_{\text{bounds}} = (0.1, 10)$ and $\alpha = 10^{-10}$. The GPR models using the Matern kernel also used $l_{\text{bounds}} = (0.1, 10)$ and $\alpha = 10^{-10}$ with a new parameter, ν , being set to 1.5. Figure 3.2 visually illustrates the differences between using the RationalQuadratic and Matern kernel to solve the same problem. Both the RationalQuadratic and Matern kernel create GPR models that have their uncertainty intervals capture the true steam flow. While there wasn't a drastic difference between the GPR models fitted for well 2 when using a RationalQuadratic or Matern kernel, there were large differences observed for well 1. Taking the 25th and 75th quantile of variance for well 1 predictions across GPR models fitted gives $\text{Quantile}_{\text{RationalQuadratic}} = (0.44, 0.56)$ and $\text{Quantile}_{\text{Matern}} = (4.39, 17.95)$. Repeating this for well 2 gives $\text{Quantile}_{\text{RationalQuadratic}} = (1.48, 10.93)$ and $\text{Quantile}_{\text{Matern}} = (0.95, 11.32)$. There is little difference in the variance quantiles when using either a RationalQuadratic or Matern kernel for well 2. However, there is a large difference when using a Matern kernel over a RationalQuadratic kernel as the well 1 variance quantile has drastically increased. This suggests that the Matern kernel produces GPR models with larger uncertainty intervals, which is desirable because the model needs to capture various possible steam flows for missing data through its uncertainty intervals. As such, even though the Matern kernel performed worse, its ability to create wider uncertainty intervals offers more flexibility and confidence when dealing with extremely sparse data and, therefore, is the kernel used for future GPR implementations using Scikit-Learn.

The geothermal operator or whoever has interest could draw realizations for steam flow for both wells across the 12-month period. These drawn realizations could then be converted back to mass flow by dividing through by dryness and solving for the missing mass flow values. A key constraint that needs to be satisfied is that the sum of steam flows across the two wells at a given time point would sum exactly to the total steam flow at the same time point. Equation (3.1.3) represents the summative total constraint for an arbitrary number of wells as

$$\text{Total}_t = \sum_{i=1}^n S_{\text{Well}_i, t} \quad (3.1.3)$$

where $S_{\text{Well}_i, t}$ is the steam flow for the i th well at time t . Conditioning is a logical solution to enforcing this constraint and is rightfully discussed in the subsequent section.

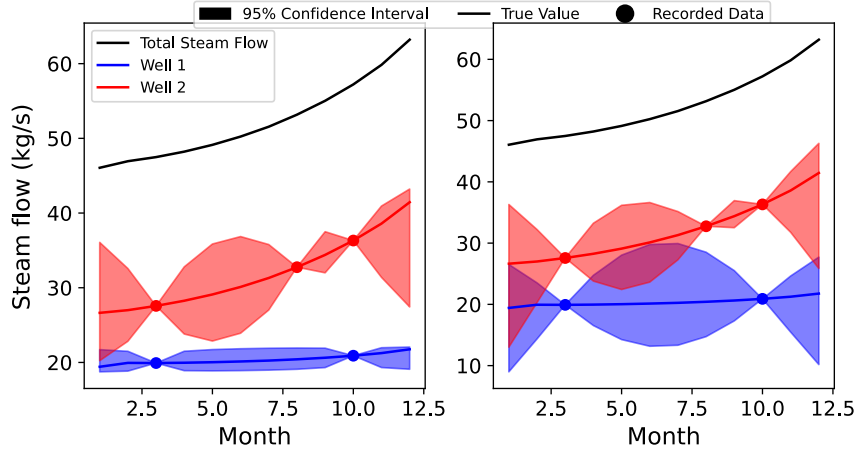


Figure 3.2: (Left) Two independent GPR models for two wells always flowing over 12 months using a Rational Quadratic kernel. (Right) Two independent GPR models for two wells always flowing over 12 months using a Matern kernel

3.2 GPR Processes with Conditioning

Using equation (3.1.3), the previous two-well problem can be resolved. Scikit-Learn doesn't have the ability to formulate GPR models under a given constraint. Instead, a semi-manual approach is implemented when solving this conditioning problem.

Consider equation (3.1.3) for the two-well example. The equation can be written out for the two-well case and then rearranged such that the subject is now a particular well, which is given by

$$S_{\text{Well}_{i,t}} = \text{Total}_t - S_{\text{Well}_{j,t}} \quad (3.2.1)$$

where $S_{\text{Well}_{i,t}}$ is the steam flow for the i th well at time t and $S_{\text{Well}_{j,t}}$ is the steam flow for the j th well at time t . In the two-well case, $i, j \in \{1, 2\}$, where $i \neq j$.

Therefore, the GPR model can be refitted as before but for time points where the steam flow is known for one well but not the other, the exact steam flow for the unknown well can be deduced at the time point. This clips the uncertainty interval for both GPR models, allowing for GPR models to still succeed with even less data. Once fitted, the mean predictions for steam flows are summed together to find the total mean steam flow predicted across both wells. The difference between this total and the true total steam flow is then calculated and both GPR model mean predictions are adjusted by this amount. Mathematically, this is given by the general expression

$$\text{MP}_{\text{Well}_{i,t}} = \frac{\text{TS}_t - \text{MP}_{\text{Sum}_t}}{n} + \text{MP}_{\text{Well}_{i,t}} \quad (3.2.2)$$

where $\text{MP}_{\text{Well}_{i,t}}$ is the mean prediction for the i th well and time t , TS_t is the true total steam flow at time t , MP_{Sum_t} is the sum of mean predictions at time t . and n is the number of wells.

The newly fitted GPR models using this conditioning method is shown in Figure 3.3. The conditioned GPR models show improvement over the previous GPR models as now extra information about wells is incorporated during fitting. This is indicated by the clipped uncertainty intervals for both wells at time points where at least one well's steam flow is known. Realizations of the steam flow across the entire time period for both wells can be drawn. This is done in a manual yet sensible way where a realization is drawn for well 1 across the time period, and then the subsequent realization is drawn for well 2 while

enforcing the constraint given by equation (3.1.3). The realizations drawn are for steam flow, meaning they must be divided through by the corresponding dryness to obtain the mass flow.

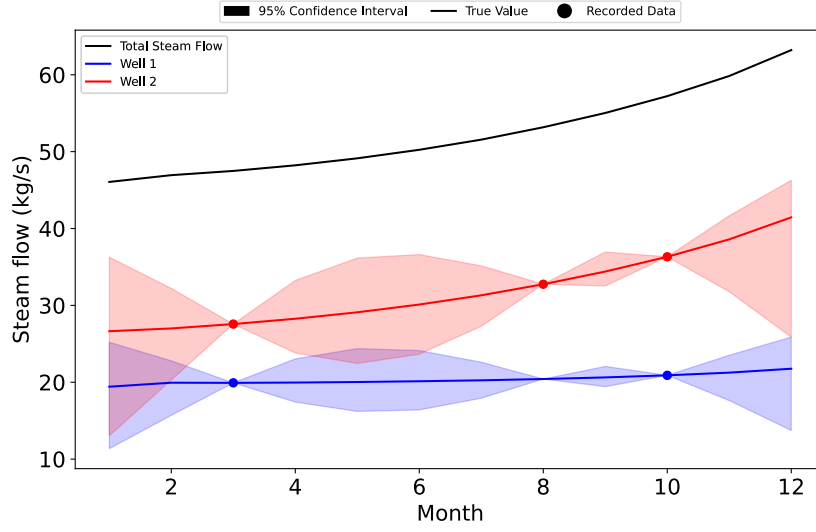


Figure 3.3: Two conditioned GPR models for two wells that are always flowing over 12 months

Figure 3.4 demonstrates a single steam flow realization being drawn across the 12-month period for both wells 1 and 2 with the equivalent mass flow realizations.

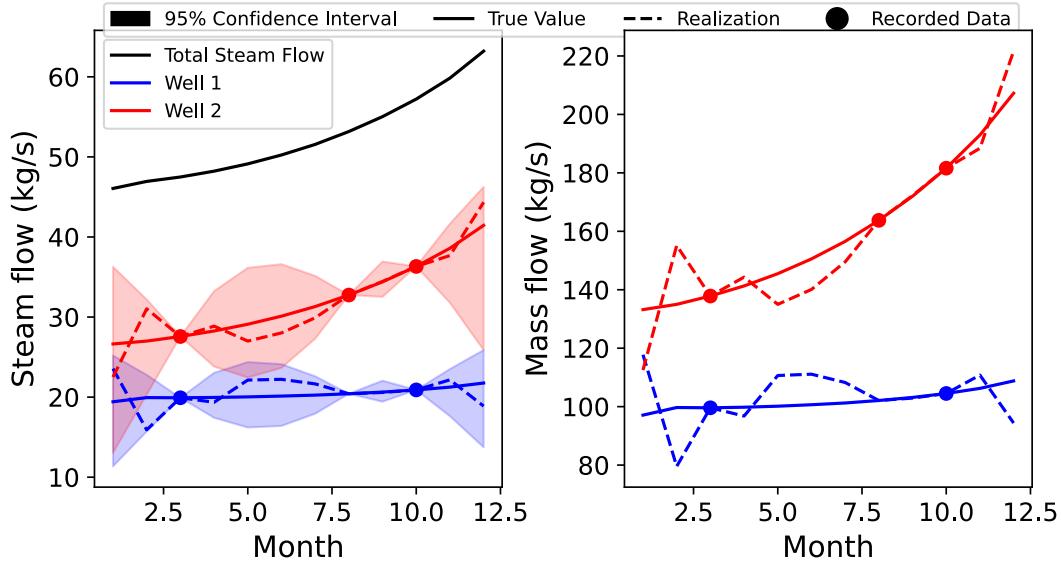


Figure 3.4: (Left) One sampled steam flow realization across 12 months for wells 1 and 2 using conditioned GPR models. (Right) The corresponding mass flow realization for wells 1 and 2 across 12 months

From Figure 3.4, the left plot demonstrates that the realizations drawn are bounded within the uncertainty intervals created. True steam and mass flows are known across all periods only because the synthetic data was generated through simulation. Real-geothermal data problems will have these true quantities replaced by expert estimates, as seen later in the report. When the steam flow realizations are converted back to mass flow, which is given in the right plot, there is a mix of result quality. When the steam flow for either well is known, the corresponding mass flow prediction matches the true mass flow exactly.

Otherwise, the mass flow realizations range in quality with $e = (0.22, 20.71)$ where e is the residual between the mass flow realization and true mass flow.

These results prove that conditioning in GPR models works and successfully enforces constraint (3.1.3). However, this method of conditioning using Scikit-Learn is undoubtedly challenging to scale for more than 2 GPR models. The ccandu library, developed by the University of Auckland modelling team, is another package that could do this conditioning process. As the package is specifically designed for Gaussian processes and conditioning methods, it's likely the results would yield more potential.

3.2.1 Improved Conditioning Using the ccandu Library

The same problem is solved using the ccandu library where there are two wells in the geothermal field and they both remain on for the entire time period. Setting up the conditioning problem is a more manual process than in Scikit-Learn but offers more control and flexibility than Scikit-Learn. A prior mean and covariance matrix is defined for each individual GPR model that needs to be fitted. Then, the prior model is created by using the total steam flow and the previously fit priors. Next, the posterior distribution is defined by using the data observations given. Combining the prior and posterior distribution allows for posterior samples to be drawn for each GPR model fitted. The sum of these posterior samples across the GPR models fitted naturally sum to the total steam flow, enforcing constraint (3.1.3). Figure 3.5 demonstrates using ccandu to solve this two well that's always on problem.

Figure 3.5 demonstrates more robust samples being drawn throughout the entire time period, except for the extrapolation period. Compared to Figure 3.3, the intervals are fairly similar and, hence, are still narrow enough to create reasonable realizations. The extrapolation period sees samples drawn with a similar uncertainty interval but behave worse as well 1 predictions increase when they are expected to remain rather flat, and well 2 predictions dip when they were expected to continue rising.

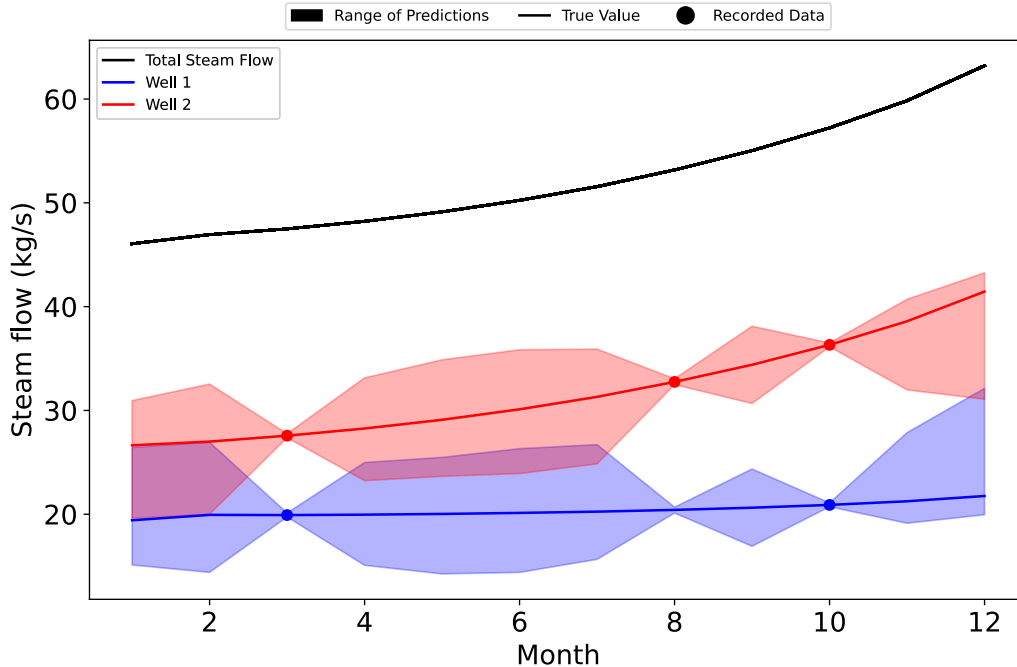


Figure 3.5: Two conditioned GPR models for two wells that are always flowing over 12 months using ccandu

As seen previously in Figure 3.4, relevant posterior samples for steam flow can be drawn and readily transformed back to mass flow. Figure 3.6 displays an example of using

ccandu to generate a steam flow realization for both wells across the time period and then converting it back to mass flow.

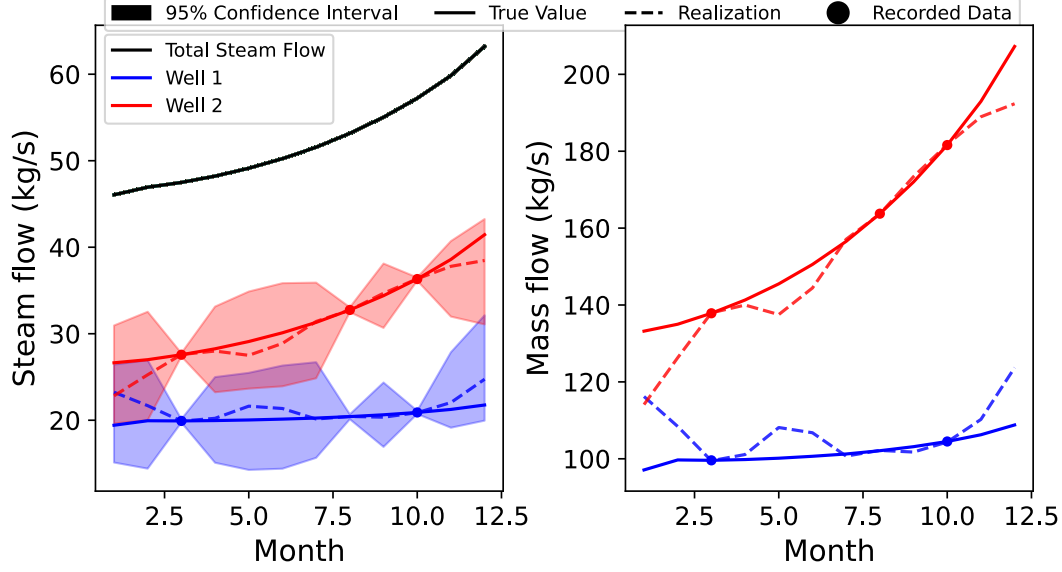


Figure 3.6: (Left) One sampled steam flow realization across 12 months for wells 1 and 2 using ccandu conditioned GPR models. (Right) The corresponding mass flow realization for wells 1 and 2 across 12 months

It has been demonstrated that GPR models can be fitted using both the ccandu and Scikit-Learn library such that steam flow realizations, and consequently mass flow realizations, can be drawn from them. A proof of concept has been generated using both the ccandu and Scikit-Learn library. Relaxing further assumptions allows for the GPR model to solve more complex problems, and is discussed in more detail in the subsequent section.

3.3 GPR with Further Relaxed Assumptions

Previously, the problem solved only considered two wells in the geothermal field and assumed these wells were always flowing. Relaxing further assumptions brings on the challenge of solving problems where multiple wells exist in the geothermal field, and these wells can be on or off. This section investigates how to solve these problems using the ccandu library, as the ccandu library proved to be a robust way of enforcing realization sums and drawing valid posterior realizations.

Consider an expansion of the previous problem where there are five wells in the geothermal field and they turn on at different times. The same process as before can be applied where two GPR models are fitted. This works by selecting a target well and then grouping all the remaining wells into a lumped well. Mathematically, it's the same as equation (3.1.3), but for this problem can be written as

$$\text{Total}_t = S_{\text{Well}_{i,t}} + S_{\text{Others}_t} \quad (3.3.1)$$

where S_{Others_t} is the steam flow across all the wells excluding the i th well at time t . S_{Others_t} can be written as

$$S_{\text{Others}_t} = \sum_{\substack{j=1 \\ j \neq i}}^n S_{\text{well}_{j,t}} \quad (3.3.2)$$

which finds the sum of steam flow across all the wells in the geothermal field except the i th one at time t , as this is the target. Suppose that in the five well geothermal field there

is interest in determining the steam flow, and thus mass flow, across well 1, then applying the processes described earlier results in Figure 3.7. Figure 3.7 demonstrates the ability to generate steam flow realizations that are fairly accurate between known periods of steam flow recordings. The prediction intervals generated for the well 1 and other well steam flows up to and including month 30 comfortably capture the true steam flow values. As alluded to before, there is a struggle to capture accurate trends after the last recorded data point and is evident here with both GPR models failing to have the prediction intervals capture the true steam flow.

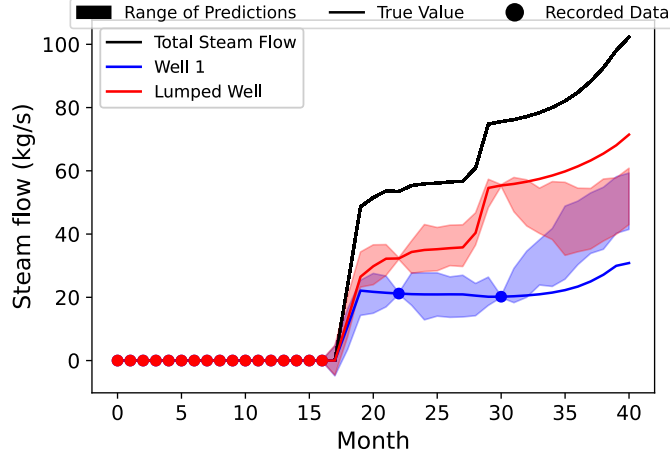


Figure 3.7: Two ccandu conditioned GPR models for five wells where well 1 is the target and the other wells are lumped into a single well. Wells can switch between on and off states and are modelled over 41 months

A geothermal operator could use this setup very effectively to interpolate steam flow values for the relevant geothermal well. By giving a target well, two GPR models would be fitted and the GPR model fit for the target well would generate a reasonable steam flow prediction, assuming no extrapolated predictions are used, that could easily be converted to the mass flow using dryness. This method is suited for inferring the detailed history of a particular well. However, it is undeniably limited when trying to infer the history of many wells, which is a common problem. This approach only helps gain insight into one well as the lumped well does not provide much value. Fortunately, this approach can be extended through the introduction of fitting GPR models for each individual well in the geothermal field.

3.4 Fitting GPR Models to Each Well

Up until now, every problem has been solved by fitting two GPR models. If there are two wells in the geothermal field, then a GPR model is fitted for each well; otherwise, all non-target wells are lumped into a single well, and a GPR model is fitted to this lumped well and target well. While challenging, a GPR model can be fit for each well in the geothermal field. This adds robustness as each well is treated separately and the geothermal operator can obtain predictions for all wells simultaneously.

Solving the same five-well problem but now fitting a GPR model for each individual well results in Figure 3.8 where five separate GPR models have been fitted to each of these five wells. There are mixed results using this method as some uncertainty intervals produced successfully capture the true steam flow behaviour within but others don't. Specific examples of this from Figure 3.8 are seen from wells 3 and 5. The well 3 uncertainty intervals for steam flow between months 18 and 21 fail to capture the true steam flow values within this interval. Likewise, the uncertainty intervals for well 5 between months 26 and 29 also fail to capture the true steam flow values within this interval.

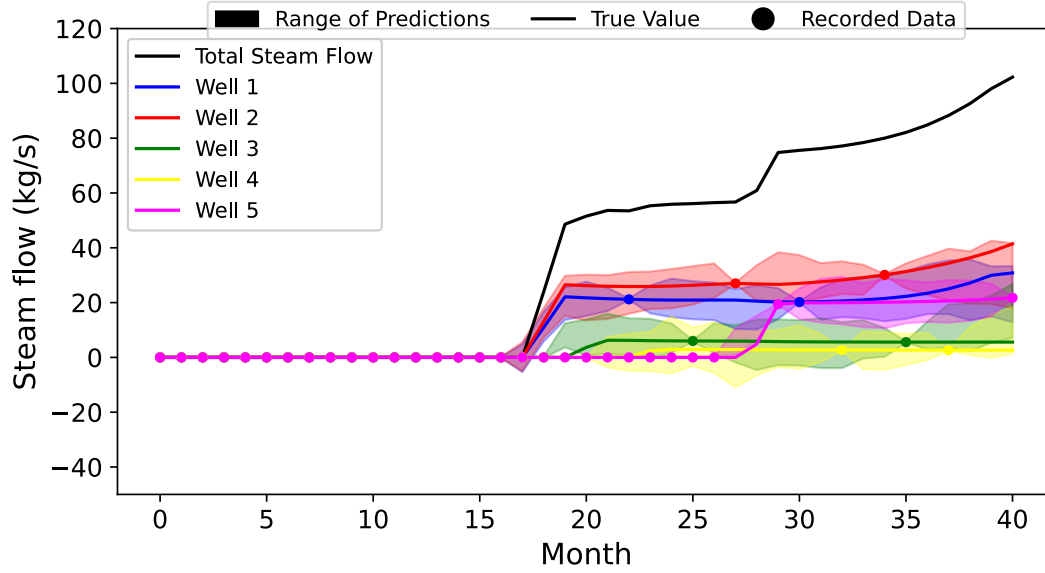


Figure 3.8: Five ccandu conditioned GPR models for five wells that can switch on and off modelled over 41 months

This section has demonstrated an effective way to solve the sparse data problem using individual GPR models fitted for each well under given constraints. While the results have been mixed, there are elements to this approach that are favourable over the lumped GPR model approach previously described, but this will be discussed in detail later. The GPR models for all 5 wells performed poorly with extrapolation, showing little improvement over the lumped GPR model approach.

3.5 Using GPR on Real-Geothermal Data

Real-geothermal data is much more challenging to work with compared to synthetic data. Often, data collected from geothermal fields are more sparse than synthetic data, have the potential to be erroneous, and are supplied in raw formats that require prior processing. Contact Energy has supplied a broad range of geothermal data collected from one of its geothermal fields. This supplied data contains the recorded total mass flow as opposed to the total steam flow, meaning there's no need to convert steam flow back to mass flow, and all equations defined previously can be applied interchangeably. Individual mass flow measurements are also sometimes reviewed by reservoir engineers to ensure their validity and to check for potential issues.

GPR models were used to generate predictions for missing mass flows for the collected data supplied by Contact Energy. Data from a separator containing four wells spanning over 1000 days was used as a subset of the total collected data. In this run, individual mass flow measurements were weighted and thus treated equally regardless of whether they were measured by a geothermal reservoir engineer or not. Applying the methods seen in section 3.4 to this collected data results in Figure 3.9 below. The first observation from Figure 3.9 is the enhanced complexity of the separator behaviour. The total mass flow experiences sudden spikes and drops and has no clear trend, a clear contrast to what has been observed using synthetic data. The time period to model for is also significantly higher than what was done for before with 1000 points now being modeled, instead of 40 using the synthetic data. The complex interactions of the wells within this separator lead to a mixture of performance across different wells at different times. One of the challenges with validating the accuracy of the GPR predictions is that the expert estimate, indicated by the solid lines in Figure 3.9, is not necessarily the true mass flow for the individual wells. Rather, it's an estimate made by an expert in the geothermal space. As such, the GPR model performance is validated against results that aren't guaranteed to be right, resulting, albeit unlikely, in the case where both the GPR predictions and expert estimates

are far off from the truth. The prediction intervals across all four wells contain the expert's estimate for a significant portion of the 1000-day period. However, the intervals are wide, as the average prediction interval width is 18 kg/s when considering wells 1, 2, and 3. As such, this interval may be considered too wide if a small number of posterior samples are drawn from this. At around day 900, well 2 switches off while well 4 switches on. The GPR models for both these wells capture the change in well status as the well 2 predictions immediately become 0 while well 4 predictions exceed 0 when they switch their status.

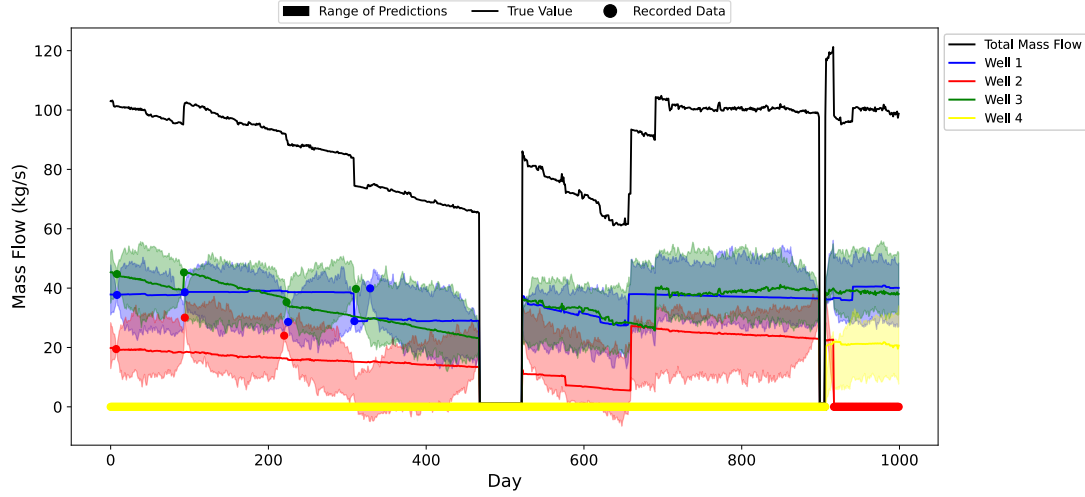


Figure 3.9: Four ccandu conditioned GPR models for a separator containing four real geothermal wells modelled over 1000 days

To facilitate confidence in these GPR models, a section of the time period is clipped to demonstrate the uncertainty interval narrowing through the introduction of measured mass flows for individual wells. Figure 3.10 below highlights a key interval within the 1000 days where there are mass flow values for wells 1, 2 and 3. The uncertainty interval becomes clipped here, meaning the GPR models consider these data points when generating predictions. Consequently, there is sufficient evidence of the GPR models performing to their expectations.

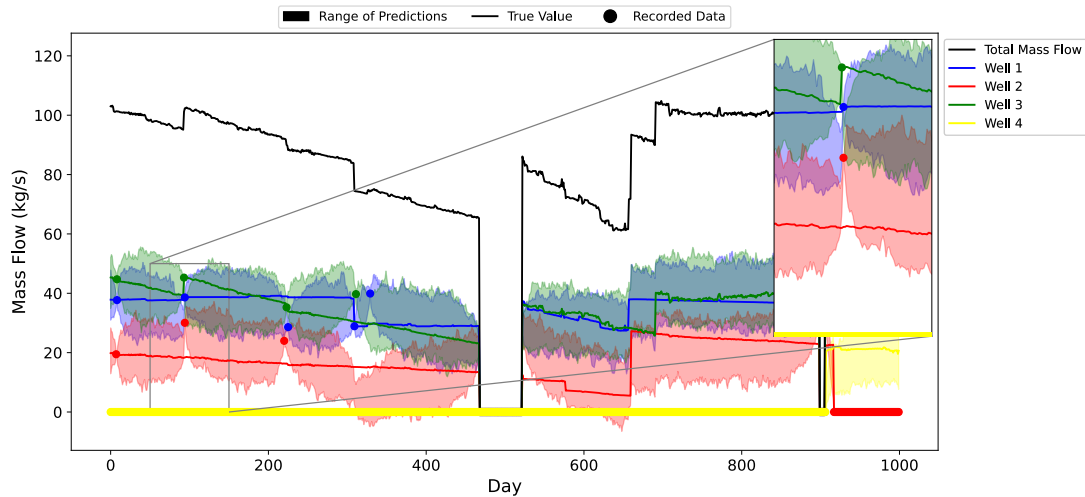


Figure 3.10: Four ccandu conditioned GPR models for a separator containing four real geothermal wells modelled over 1000 days with the zoomed section between days 50 and 150

As mentioned earlier, the prediction intervals produced for the wells are potentially too wide. If a small number of posterior samples was drawn, then the range of predictions

made could vary significantly. Implementing a 95% confidence interval on these GPR models produced solves this wide prediction interval problem by focusing on the range where 95% of predictions are expected to lie in. Figure 3.11 shows the effects of using a 95% confidence interval rather than the entire prediction interval used in Figures 3.9 and 3.10. The average confidence interval, when using wells 1, 2, and 3, drops to 6.1 kg/s, which is a reduction of 11.9 kg/s when using the entire prediction interval used in Figures 3.9 and 3.10. While the intervals are narrower using a 95% confidence interval, this has resulted in certain expert estimates being completely missed during the 1000-day period. For example, between days 100 to 220 the well 2 confidence intervals miss the expert’s estimates of 15 to 20 kg/s as the confidence interval doesn’t contain these values. This happens because the GPR models obey the recorded data points by fitting models, and hence generating predictions, that go through these points. As such, on day 100 when the mass flow is 30 kg/s the GPR model predictions pull up while the expert’s estimate does not because there is no constraint enforcing this.

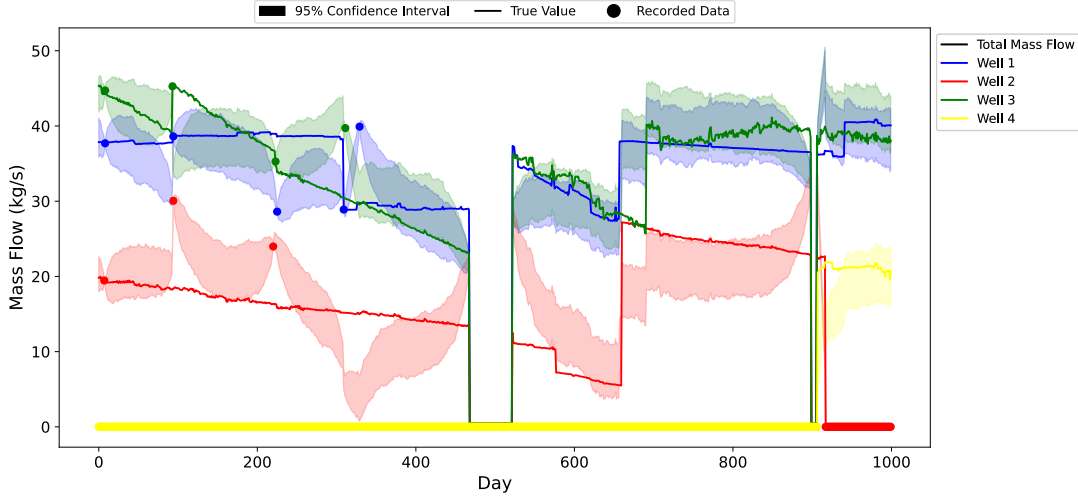


Figure 3.11: Four ccandu conditioned GPR models for a separator containing four real geothermal wells modelled over 1000 days using 95% confidence intervals rather than full prediction intervals

Using the confidence intervals obtained, the expert’s estimates, and the measured individual mass flows allows for further insight from applying GPR to solving this real-geothermal problem. Table 3.1 shows the relevant performance metrics for the GPR model on this real-geothermal application which compares the average RMSE score and average uncertainty interval width for both the GPR model using the prediction interval, Figure 3.9 and 3.10, 95% confidence interval, Figure 3.11. Across all four wells, using a 95% confidence interval gives significantly lower uncertainty intervals as, on average, intervals were 9.3069 kg/s narrower than when using prediction intervals.

Table 3.1: Comparative scores across the four wells when using prediction intervals versus 95% confidence intervals for quantifying uncertainty

Well	Average Uncertainty Interval for Prediction Interval	Average Uncertainty Interval for 95% Confidence Interval	Average RMSE Score for Prediction Interval	Average RMSE Score for 95% Confidence Interval
Well 1	18.4187	6.2561	3.5138	3.4262
Well 2	16.8419	5.7225	4.8775	4.7973
Well 3	18.8247	6.3658	3.8466	3.8401
Well 4	2.2762	0.7894	1.3455	1.3905

3.6 GPR Versus Mathematical Formulation

As mentioned earlier in the report, a comparison of the GPR method against the mathematical formulation is presented. Both approaches used the same data which was limited

to a total period of 1461 days and included three wells feeding into the separator. Figures 3.12 and 3.13 compare the GPR model approach against the mathematical formulation approach to the problem described. Comparatively, neither method is clearly better, but both methods deliver better results in certain aspects. The GPR method finds more success in reaching the expert's estimates for the higher well 2 values observed between days 400 to 600. The GPR method also makes use of the recorded data points for individual wells effectively as the models pass through these known points. This contrasts with the mathematical formulation approach as it fails to go near certain data points, which happens because the polynomial tries to go through points as closely as possible but emphasises the smoothness of the overall function to prevent oscillations. For example, on approximately day 570, the GPR model for well 2 goes through this data point, allowing it to align with the expert's prediction. At the same time, the mathematical formulation ignores this point. The mathematical formulation does have clear areas of enhanced performance over the GPR method as, for example, predictions made for well 1 better align with the expert's estimates, whereas the GPR estimates, especially between days 600 and 700, suddenly pull up when there was little reason to do so. These predictions suddenly change to meet the total flow requirement that is strictly enforced. Predictions made using the mathematical formulation method are all positive too, while the GPR predictions sometimes are negative, so the predictions by this other method may be considered more robust in certain areas.

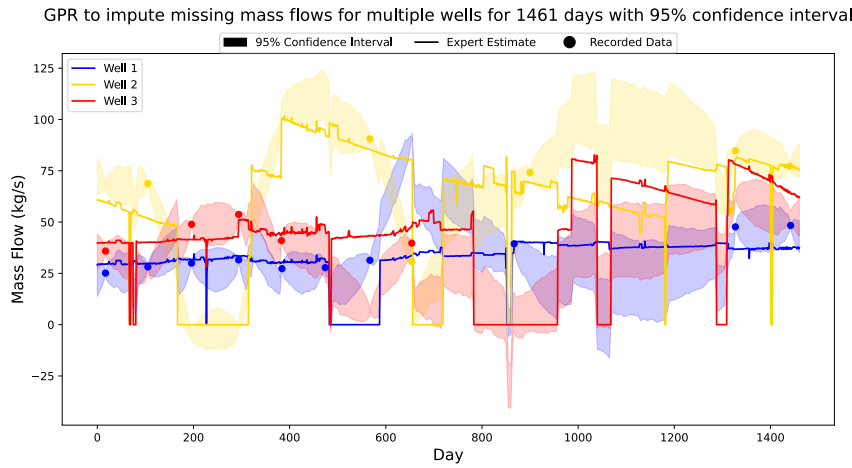


Figure 3.12: Three ccandu conditioned GPR models for a separator containing three real geothermal wells modelled over 1461 days

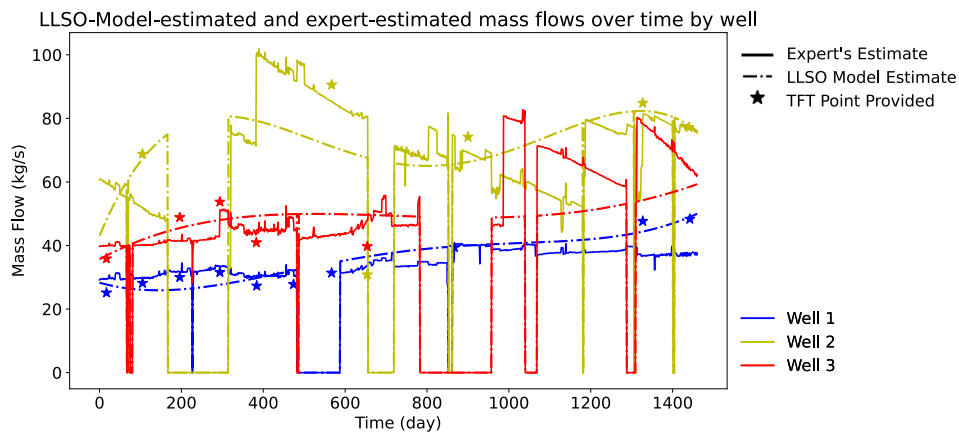


Figure 3.13: Mathematical formulation approach on real-geothermal data for a separator containing three wells modelled over of 1461 days

Chapter 4

Discussion

The results of GPR on various problems have been discussed through detailed analysis of the performance of GPR on various synthetic datasets and real-world problems. This section outlines the core findings of using GPR to solve this sparse data problem. Suggestions for future work, as well as limitations, are thoroughly discussed to encourage further thought into this approach.

4.1 Findings

The research undertaken has demonstrated the feasibility of using GPR to impute missing mass flows for both synthetic and real-life problems. GPR has yielded excellent results in estimating missing mass flows for the simpler synthetic problems seen in sections 3.1, 3.2, and 3.2.1 as the uncertainty intervals across the entire time periods captured the true value. While the problem solved was simple, the GPR models produced were robust, which justifies GPR as a tool that is suited to solving this problem. Even with more complex synthetic data, as seen in sections 3.3 and 3.4, the performance of GPR was still good with the uncertainty intervals mostly capturing the true values. Exceptions to this were observed for extrapolated predictions, but this is discussed later as a limitation as sometimes the trend changed suddenly, and the true value was not captured inside the uncertainty interval. Regardless, the performance of GPR across all synthetic problems is good as predictions were generally robust. Applying GPR to real-geothermal data resulted in worse performance when compared to synthetic data, but still reasonable estimates at certain times. As seen in section 3.5, the enhanced complexity of the real data caused the GPR models to oscillate frequently, sometimes failing to capture the expert's estimates for mass flow within their uncertainty intervals. There are cases where the expert's estimate fails to go through the recorded data when theoretically it should. As such, this explains why sometimes the expert's estimate and GPR models differ so much as the GPR models must be fitted through these points while there appears to be no such constraint for the expert's models. Experts need to make decisions regarding the importance of the data points, so they are likely to assign a non-equal weighting to them, encouraging their models to go through certain data points and avoiding others. Currently, the GPR model uses an equal weighting scheme where every point is equally important. Across all GPR models implemented using Scikit-Learn, the average RMSE score when comparing the mean GPR model predictions to the true values was 1.3868 kg/s and the average uncertainty interval width was 5.9376 kg/s. While the RMSE score was small and the uncertainty interval was narrow, these scores only reflect the simple synthetic cases that would unlikely be observed in real-geothermal contexts. A more representative expected RMSE error and uncertainty interval comes from the GPR models fit on the real-geothermal 1000-day problem, visualized in Figures 3.9 and 3.10. Using Table 3.1, the average RMSE score, when excluding well 4 as it is off for the majority of this period, is 4.0793 kg/s, and the average uncertainty interval is 18 kg/s. However, as seen and discussed in section 3.5, using the 95% confidence interval rather than the entire prediction interval decreased the average uncertainty interval to 6.1 kg/s while reducing the average RMSE to 4.0212 kg/s.

The reduction in error wasn't significant but the reduction in uncertainty interval was. As such, using a 95% confidence interval rather than the entire prediction interval is preferable when making predictions near the prediction interval mean as the RMSE score remains roughly similar with significantly lower uncertainty. As demonstrated in Sections 3.3 and 3.4, there is little reason to lump wells into a single well as opposed to fit a GPR model for each well. Both the lumped well implementation and separate well implementation solved the synthetic data problem where flow could switch on and off and was modelled over 41 months. When using the separate well approach instead of the lumped well approach, the average RMSE reduced by 5.2272 kg/s and the average uncertainty interval reduced by 0.9122 kg/s. The lumped well approach also is less flexible as it only provides useful predictions for one well. As such, the worse performance and reduced flexibility of the lumped well approach means it should be avoided as it provides little benefit relative to the separate well approach.

4.1.1 Limitations

The quality of data provided for individual wells by geothermal operators influences the quality of predictions made by GPR models. While it's assumed the data provided will be sparse, there is also an assumption the data is not erroneous, and if it is, it has been reviewed by a reservoir engineer. If erroneous data is supplied, it shouldn't be used because the GPR models place a large emphasis on creating models that go through these points. If used, the predictions made may be invalid because they will be making predictions using erroneous data. The collected data requires expert assessment by reservoir engineers to determine the quality. Reservoir engineers can then give weightings for these data points, which determine the influence of each data point.

GPR and most mathematical methods are unreliable for extrapolation. The model cannot use trends and correlations previously used as there is no assumption they continue. As such, GPR can't make informed predictions for this unobserved period, which often leads to unexpected results, such as predictions suddenly rising or dropping. Care should be taken when interpreting results after the last measured data point.

Drawing from the previous two limitations mentioned, even though GPR is well-suited for sparse data problems, there still needs to exist two data points for each individual well at the minimum. This is because correlations created between points drive predictions, and if only one point exists, there can be no correlation drawn for just an individual point. If GPR is applied to a well with only one recorded data point, there is no expectation of quality predictions.

4.2 Suggestions for Future Work

Currently, there are no restrictions on whether the posterior samples drawn using the GPR model are positive or negative. Enforcing a non-negativity constraint on this current formulation would prevent any negative value samples from being drawn, which in this context are not valid values, and so help improve the robustness of the current GPR models. This non-negativity constraint would be beneficial to the quality of the model as it would remove negative values and prevent values from other wells from jumping upwards to compensate for this negative value. Geothermal operators and reservoir modellers would benefit from this because there would be no need to worry about invalid samples.

Creating a custom kernel is a more complex and challenging task, yet it would significantly benefit the quality of GPR models produced. Currently, the kernels the models use are common kernels such as the Matern kernel. These kernels allow for some flexibility, as the standard deviation and length scale can be defined. However, as observed in sections 3.5 and 3.6, these kernels may be too broad for these complex problems. Implementing a custom kernel that allows more flexibility between each set of data points may lead to better solutions as more accurate correlations are captured. Successful implementation of this custom kernel likely would yield better results as unwanted oscillations between data points would be reduced and thus more frequently align with the expert's estimates.

Geothermal operators would benefit from this as the solutions produced would be more robust and reliable.

The GPR models all assigned equal weightings of importance for the data collected. This ensured that the GPR models would always pass through the data points. However, giving the data different weightings based on whether it has been reviewed by a reservoir engineer and the overall quality of the data point may reduce the large oscillations observed across the GPR models. This may mean the GPR model does not go through every data point, so it is important to consider these ratings extensively.

Fitting multiple GPR models simultaneously on a given condition is a slow process. Specifically, drawing the posterior samples is the most time-consuming part, taking up to 10 minutes when drawing 200 posterior samples for 1000 time points. Currently, by selecting a subset of time points to create the posterior model, the time to draw 200 posterior samples decreases significantly. Since drawing posterior samples is a time-consuming action, rather than creating the posterior model, fitting the posterior model on all the time points and only drawing posterior samples for a subset of these time points is an effective way to produce accurate predictions in a reasonable time while ensuring flexibility. Python is an interpreted programming language which is typically easier to understand and use but is slower than compiled languages. While the code can be optimized further through minimizing time-consuming operations, using a compiled language would be required to make extremely efficient and fast code. As geothermal fields can become very large, solving this issue would be invaluable for geothermal operators as GPR predictions could be drawn in a reasonable time.

Chapter 5

Conclusions

The main objective of this research was to determine whether a GPR approach was suitable for imputing missing mass flows for individual geothermal wells. Current methods for mass flow imputation are inefficient and require expert knowledge to have quality estimates, but these sometimes might still be inaccurate. These missing mass flows are typically inputs into reservoir models, so inaccuracies translate into reservoir models containing errors. When solving synthetic data problems, GPR has consistently presented itself as a quality method that can generate fairly accurate estimates for missing steam flows that can be converted back to mass flows readily. However, when using real-life data, the enhanced complexity of the problem leads to the GPR performing worse for these real-geothermal problems than the synthetic ones, as there are periods where the GPR predictions are good, and others are inaccurate. As such, a strong foundation has been laid for using GPR as a method for imputing missing mass flows for individual wells, and a proof of concept was established using synthetic data. Moving forward, improvements can be made to the current GPR method to make predictions more robust and increase the viability of this method for use in the industry. Enforcing a non-negativity constraint would improve the robustness and reliability of this method as no invalid samples would be drawn. Exploring the potential of a custom kernel can be considered grounds for future studies where the implementation and performance of these custom kernels can be explored in-depth. Assigning weights to the data points may be an appropriate way forward to reduce the oscillations observed by the GPR models as the models may no longer need to go through every data point. Improving the efficiency and speed of GPR by effective posterior sampling would make this method much more attractive for geothermal operators, as predictions could be made in reasonable time frames. GPR has mostly met its objective of providing an effective way of imputing missing mass flows for individual wells. The proof of concept developed has confirmed the feasibility of using GPR to impute these missing mass flows. While more success was observed when using synthetic data rather than real-life data, enforcing the non-negativity constraint and utilizing custom kernels may significantly enhance the existing models developed for solving these real-life data problems. Further research into using GPR for mass flow imputation would be valuable to geothermal operators and reservoir modellers as they could have robust estimates for how much mass was taken from a well at a given time, assuming this isn't an extrapolated value. Geothermal reservoir modellers use these mass flows as inputs to their models, so knowing accurate mass flows across wells would allow them to model everything accurately. These accurate models could be used to determine the overall health of the geothermal reservoir and inform the geothermal operator about safe fluid extraction rates for these wells.

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